Q4B: Calculate the relativistic kinetic energy of a particle with a given velocity.

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Relativistic Kinetic Energy of a Particle

Setup

Let a particle of rest mass m_0 move with velocity v relative to an inertial frame. We wish to derive and calculate its relativistic kinetic energy.

Key Definitions

The **Lorentz factor** is defined as:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

where $c = 3 \times 10^8$ m/s is the speed of light in vacuum.

Derivation

The **relativistic total energy** of the particle is:

$$E = \gamma m_0 c^2$$

The **rest energy** (energy when $v = 0$, i.e., $\gamma = 1$) is:

$$E_0 = m_0 c^2$$

The **relativistic kinetic energy** K is the difference between the total energy and the rest energy:

$$\boxed{K = E - E_0 = (\gamma - 1)\, m_0 c^2}$$

Explicit Form

Substituting γ:

$$K = \left(\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} - 1 \right) m_0 c^2$$

Classical Limit Verification

In the non-relativistic limit $v \ll c$, we expand γ using the binomial approximation:

$$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-1/2} \approx 1 + \frac{1}{2}\frac{v^2}{c^2} + O\left(\frac{v^4}{c^4}\right)$$

Therefore:

$$K \approx \left(1 + \frac{1}{2}\frac{v^2}{c^2} - 1\right) m_0 c^2 = \frac{1}{2} m_0 v^2$$

which recovers the classical Newtonian kinetic energy, as expected.

Numerical Example

Let $m_0 = 9.109 \times 10^{-31}$ kg (electron rest mass) and $v = 0.8c$.

Compute γ:

$$\gamma = \frac{1}{\sqrt{1 - (0.8)^2}} = \frac{1}{\sqrt{1 - 0.64}} = \frac{1}{\sqrt{0.36}} = \frac{1}{0.6} = \frac{5}{3} \approx 1.6667$$

Compute the rest energy:

$$E_0 = m_0 c^2 = (9.109 \times 10^{-31})(3 \times 10^8)^2 = 8.198 \times 10^{-14} \text{ J}$$

Compute the kinetic energy:

$$K = (\gamma - 1)\, m_0 c^2 = \left(\frac{5}{3} - 1\right)(8.198 \times 10^{-14}) = \frac{2}{3} \times 8.198 \times 10^{-14} \approx 5.465 \times 10^{-14} \text{ J}$$

Result

$$\boxed{K = (\gamma - 1)\, m_0 c^2 \approx 5.465 \times 10^{-14} \text{ J}}$$

for an electron travelling at $v = 0.8c$.